

JOEL ANDRÉS VIOLANTE MADRID

Full Stack Developer (Web & Mobile) | Geospatial Data Specialist

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PROFESSIONAL SUMMARY

Full Stack Developer with 4+ years of experience building high-performance web and mobile applications, with a specialized focus on geospatial data science. Hybrid profile combining modern software architectures (React, Angular, Flutter, Node.js, Supabase) with remote sensing and spatial analysis (Sentinel-2, Google Earth Engine, Python, PostGIS). Led national-scale projects for CONAFOR in partnership with the World Bank, delivering forest monitoring systems with offline-first mobile apps, geospatial data pipelines, and real-time interactive maps. Actively integrates AI tools (Claude, GitHub Copilot, Cursor) and multi-agent workflows to optimize development speed and code quality. Proven DevOps capabilities (AWS, Docker, CI/CD) and hands-on experience in eCommerce, SaaS platforms, and cross-platform mobile development.

KEY ACHIEVEMENTS

National Impact	Led development of SPPIF for CONAFOR/World Bank — a national-scale forest fire monitoring system with offline-first mobile apps deployed across field teams
Performance	Migrated Android/Swift native apps to Flutter, reducing development time by ~40% and unifying codebase into a single cross-platform repository
Satellite Data	Automated Sentinel-2 image processing pipeline using Python/GEE, cutting geospatial analysis time by 30% and improving data reliability
eCommerce	Built full-featured eCommerce platform with Angular + Stripe + AWS for Alpha Lubricants, expanding digital market reach by 36%
Efficiency	Implemented Python-based process automation for CONAFOR critical systems, improving operational efficiency by 50% and reducing software incidents by 75%

PROFESSIONAL EXPERIENCE

Full Stack Developer & Geospatial Data Specialist | CONAFOR / UJED / Fondo Mexicano para la Conservación de la Naturaleza, A.C. (FMCN)

Durango, Mexico • January 2026 – Present

- Continuing to lead development and evolution of SPPIF, expanding real-time analysis capabilities and adding new satellite data layers.
- Enhancing geospatial algorithms for forest fire risk detection and integrating updated Sentinel-2 multispectral data streams.
- Providing technical mentorship to development team and coordinating requirements with CONAFOR stakeholders.

Stack: Flutter Supabase PostgreSQL PostGIS Python Google Earth Engine Angular TypeScript

Full Stack Developer & Geospatial Systems Consultant | CONAFOR / UJED

Durango, Mexico • April 2025 – December 2025

- Led modernization of SPPIF with enhanced cartographic visualizations and improved accuracy of forest fire perimeter data.
- Vectorized and processed fire perimeters using Python, geemap, geopandas, and Google Earth Engine on Sentinel-2 imagery (RGB, RBR indices).
- Built advanced interactive map components in Angular/TypeScript for real-time geospatial visualization dashboards.
- Implemented backend features in PHP/HTML/CSS for internal institutional management systems.

Stack: Python geemap geopandas Google Earth Engine Sentinel-2 Angular TypeScript PHP PostGIS

Full Stack Developer & Remote Sensing Specialist | CONAFOR / UJED / World Bank

Durango, Mexico • February 2024 – March 2025

- Designed and built cross-platform geospatial mobile apps (Android → Swift → Flutter) for forest field teams, featuring offline-first sync via SQLite + SyncService.
- Architected REST backend using Supabase, PostgREST, and PostgreSQL/PostGIS, enabling seamless spatial data management and device-to-server synchronization.
- Automated satellite data pipelines (Sentinel-2, GOES, NOAA) using Python and Google Earth Engine, cutting analysis time by 30% and improving data quality.
- Developed interactive mapping components in Angular/TypeScript for SPPIF (Forest Fire Prediction & Prevention System).
- Coordinated a cross-functional development team and reported directly to CONAFOR and World Bank stakeholders.

Stack: [Flutter](#) [Android](#) [Swift](#) [Supabase](#) [PostgREST](#) [PostgreSQL](#) [PostGIS](#) [SQLite](#) [Python](#) [GEE](#) [Sentinel-2](#) [Angular](#) [TypeScript](#)

Full Stack Developer & DevOps Engineer | [Filtros y Aceites del Norte \(Alpha Lubricants\)](#)

Durango, Mexico • February 2024 – July 2024

- Built a complete eCommerce platform with Angular, Tailwind CSS, and TypeScript, integrating Stripe payment processing and an interactive product catalog.
- Provisioned cloud infrastructure on AWS (Lightsail + DynamoDB), optimizing costs and ensuring high availability.
- Implemented CI/CD pipeline with Docker and Git, standardizing development and production environments.
- Grew digital brand reach by 36% through responsive design, technical SEO, and UX improvements.

Stack: [Angular](#) [TypeScript](#) [Tailwind CSS](#) [Stripe](#) [AWS](#) [Lightsail](#) [DynamoDB](#) [Docker](#) [Git](#)

Full Stack Developer — Freelance | [Personal Project / Chaimanzana Estudio Creativo](#)

Durango, Mexico • August 2024 – December 2024

- Designed and built a SaaS platform for digital event invitations (weddings, parties), including guest management, RSVP tracking, and per-event customization.
- Delivered sub-1.5s load times via React + Tailwind CSS frontend and Next.js backend with SSR/SSG optimization.
- Configured production infrastructure on Vercel + AWS + Linux with Docker for auto-deployments and high availability.

Stack: [React](#) [Next.js](#) [Tailwind CSS](#) [Docker](#) [Vercel](#) [AWS](#) [Linux](#)

Full Stack Developer — Web & Mobile | [Advante Digital México](#)

Durango, Mexico • August 2022 – February 2024

- Delivered MySpaLive end-to-end: Flutter/Dart mobile app from scratch, CakePHP API endpoints, and JavaScript/Bootstrap admin panel for the aesthetic industry.
- Led migration of native Android (Java/Kotlin) and Swift apps to Flutter, unifying codebase and boosting system performance by 60%.
- Integrated HTTP request flows, unit testing, and MySQL data layer for real-time client and service management.
- Contributed to TRT HUB (Angular/TS): real-time analytics dashboards, REST API integration, and component optimization that improved load times and UX.
- Extended Mint (dental Flutter app) with new features and implemented custom-styled forms in the MySpaLive corporate website.

Stack: [Flutter](#) [Dart](#) [Android](#) [Swift](#) [CakePHP](#) [PHP](#) [MySQL](#) [JavaScript](#) [Bootstrap](#) [Angular](#) [TypeScript](#)

EDUCATION & TRAINING

B.Sc. in Software Engineering | [Universidad Politécnica de Durango](#)

Durango, Mexico • 2019 – 2022

Introduction to Geographic Information Systems (GIS) | [Universidad Juárez del Estado de Durango](#)

Durango, Mexico • 2023 – 2024

TECHNICAL SKILLS

Frontend	React • Next.js • Angular • Vue.js • TypeScript • JavaScript • Tailwind CSS • Bootstrap • HTML5 • CSS3
Backend	Node.js • PHP • CakePHP • Python • FastAPI • Supabase • PostgREST • RESTful APIs • GraphQL

Mobile	Flutter • Dart • Android (Java/Kotlin) • Swift • SQLite • Offline Sync • Riverpod
DevOps & Cloud	AWS (Lightsail, DynamoDB) • Docker • Vercel • Git • CI/CD • Linux • Firebase • Oracle Cloud
Data & Geospatial	Python • pandas • geopandas • geemap • PostGIS • PostgreSQL • Google Earth Engine • Sentinel-2 • QGIS • GDAL
Databases	PostgreSQL • MySQL • MongoDB • DynamoDB • SQLite • PostGIS
Methodologies	Agile / SCRUM • Kanban • DevOps • TDD • REST / GraphQL
AI Dev Tools	Claude • GitHub Copilot • Cursor • AI Agents • Prompt Engineering • LLM Orchestration

AI-AUGMENTED DEVELOPMENT

Actively integrates AI tooling into the software development lifecycle to accelerate delivery, improve code quality, and stay at the forefront of modern engineering practices.

- Orchestrates specialized AI agents (Claude, GitHub Copilot, Cursor) to handle discrete tasks — code generation, refactoring, unit test creation, and automated documentation — within a structured multi-agent workflow.
- Applies proven Prompt Engineering best practices to produce reliable, consistent, and reusable outputs across real development contexts.
- Leverages AI-assisted code review and intelligent debugging to cut iteration cycles, maintain high code quality, and reduce time-to-delivery on complex features.
- Follows responsible AI-assisted development principles: validating every AI-generated output, maintaining full ownership of architecture decisions, and using AI as a force multiplier rather than a replacement for engineering judgment.

Tools: [Claude \(Anthropic\)](#) [GitHub Copilot](#) [Cursor IDE](#) [AI Agents](#) [LLM Orchestration](#) [Prompt Engineering](#)

LANGUAGES

Spanish: Native • **English:** Upper-Intermediate (technical documentation, professional communication, code review)